

Yiwei Shu

Ph.D. Student in Robotics | Field and Planetary Robotics

📍 Beijing, China

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Research Profile

- Ph.D. student in robotics at the University of Chinese Academy of Sciences and the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences. Research interests include reliable autonomous exploration for field and planetary robots, active SLAM, risk-aware motion planning, reactive manipulation, and low-cost aerospace robotic systems. Open to visiting Ph.D. opportunities and international research collaborations.

Education

Ph.D.	University of Chinese Academy of Sciences , Robotics and Space Robotics • Affiliated with the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences. • Research on autonomous exploration, legged field robots, active SLAM, motion planning, and planetary robotics.	Beijing, China 2024 – present
B.Eng.	ShanghaiTech University , Electronic Information Engineering • Undergraduate research in robotics, visual control, point-cloud registration, medical imaging, and integrated-circuit design.	Shanghai, China Sept 2020 – June 2024
Exchange	University of Illinois Urbana-Champaign , Exchange Program • Completed Quantum Physics, Data Science Discovery, and Statistical Analysis with A+ grades.	Urbana, IL, USA Jan 2023 – May 2023

Experience

Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences , Ph.D. Researcher University of Chinese Academy of Sciences • Develop reliable autonomy for Unitree Go2 quadruped robots operating in uncertain, unstructured, and lunar-analogue environments. • Build simulation and evaluation pipelines for active SLAM, world-model reliability, terrain traversability, and sim-to-real navigation. • Research reactive whole-arm manipulation, grasp planning, risk-aware motion planning, and autonomous aerospace control.	Beijing, China 2024 – present
ShanghaiTech Automation and Robotics Center (STAR Center) , Undergraduate Researcher • Developed fixed-wing guidance and visual-control simulations for accurate flight and target landing under disturbances. • Reconstructed 3D scenes from LiDAR point clouds and implemented ICP-based point-cloud registration using PCL. • Connected aerodynamic simulation, control-model development, and physical launch experiments in a sim-to-real workflow.	Shanghai, China Sept 2022 – June 2024

Manuscripts Under Review and Submitted

Finite-State Visual Servo Control for Autonomous Target Landing of Low-Cost Underactuated Fixed-Wing Aircraft Yiwei Shu , Shuquan Wang. Manuscript under review at <i>Aerospace Science and Technology</i> . Proposed an IMU/monocular-camera finite-state visual servo controller and validated it with CFD, randomized simulations, and real flight tests; reduced final image-plane radial error from 0.5049 to 0.0027 against a sliding-mode baseline.	June 2026
From End-Effector Targets to Whole-Arm Postures: Configuration-Space Waypoint Recovery for Reactive Manipulation Yiwei Shu et al. Manuscript submitted to <i>IEEE Robotics and Automation Letters</i> (submission status: received). Introduced PostureWaypoint for online configuration-space recovery; evaluated four manipulators across six constrained 3D tasks and validated the method on an RM75 robot.	July 2026
RKILPlanner: Risk-based Kinematic Imitation Learning Planning Towards Human-Like Driving in Highway Scenarios Yiwei Shu , Shuquan Wang. Manuscript submitted to <i>ROBIO 2026</i> . Combined Driver's Risk Field maps, driver-style encoding, and LSTM/CNN features in a differentiable kinematic imitation-learning planner evaluated on the HighD naturalistic highway dataset.	2026

Publications and Presentations

Expanding Grasp Reachability: Field-based Pose-constrained Two-stage Grasp Planning

July 2026

Yiwei Shu, Shuquan Wang, Jun Li, Hanying Sang, and Yidong Ye. Accepted full paper and oral presentation at *ICoSR 2026*. Developed reachability and manipulability fields with a two-stage policy that expands pose-constrained grasping near workspace boundaries; validated in simulation and on a 7-DoF RM75 manipulator.

Learning-based Extended-workspace Manipulation: A Self-relocating Space Crawler for Autonomous Deep-space Outposts

2026

Yiwei Shu, Shuquan Wang. Oral presentation at the *46th COSPAR Scientific Assembly 2026*. Proposed a passive-rail, self-relocating space crawler and hierarchical reinforcement-learning framework for autonomous extended-workspace manipulation in deep-space outposts.

Attitude Control of Unpowered Aircraft in Flight Based on Kalman Filter Visual Servo-PID Optimization Algorithm

July 2025

Yiwei Shu, Shuquan Wang. Accepted for presentation and publication in the proceedings of the *44th Chinese Control Conference (CCC 2025)*. Combined visual servoing, adaptive PID control, and Kalman filtering for robust low-cost unpowered-aircraft attitude control.

Projects

Reliable Autonomous Exploration for Unitree Go2

2026 – present

Reliable active SLAM and navigation for quadruped robots in lunar-analogue and unstructured environments.

- Integrate LiDAR, depth sensing, grid-based mapping, and exploration planning for the Unitree Go2 platform.
- Build Isaac Lab lunar-terrain benchmarks and real-physics evaluation pipelines for legged navigation.
- Investigate calibrated world models, distribution shift, terrain traversability, and safe sim-to-real deployment.

PostureWaypoint for Reactive Whole-Arm Manipulation

2025 – present

Configuration-space waypoint recovery for online manipulation in cluttered and constrained workspaces.

- Resolve end-effector waypoint ambiguity through whole-arm clearance, orientation consistency, and joint-feasibility constraints.
- Evaluate on RM75, Franka Panda, UR5e, and KUKA iiwa manipulators with simulation and real-robot experiments.

Field-based Pose-constrained Two-stage Grasp Planning

2025 – 2026

Reachability- and manipulability-aware grasp planning for objects near manipulator workspace boundaries.

- Construct reachability and manipulability fields directly from the robot model.
- Use a sweep-then-pick policy when direct grasping is infeasible under strict pose constraints.

Autonomous Fixed-Wing Target Landing

2023 – 2026

Low-cost visual-servo control for underactuated and unpowered fixed-wing aircraft.

- Combine CFD-based aerodynamic characterization, IMU guidance, monocular visual servoing, and adaptive control.
- Validate controllers through randomized simulation and physical flight experiments.

Technical Skills

Programming: Python, MATLAB, C, C++, R, Verilog, LaTeX

Robotics: ROS/ROS 2, PCL, Unitree Go2, RM75, STM32, LiDAR and depth sensing

Simulation and Learning: Isaac Lab, MuJoCo, PyTorch, scikit-learn, numerical simulation

Engineering Tools: Git, Linux, SolidWorks, Altium Designer, ANSYS Fluent, Cadence, Vivado

Selected Honors

- Second Prize, East China Division, 6th National Undergraduate IC Innovation and Entrepreneurship Competition (2022)
- Second Prize, RoboMaster Eastern Division (2022)
- Third Prize, RoboMaster National Competition (2022)

Teaching and Leadership

- Teaching Assistant, Digital Circuits, ShanghaiTech University (September-December 2022): led tutorials and assisted with grading.
- Sentry Team Lead, RoboMaster (2021-2023): coordinated embedded programming, PCB design, mechanical modeling, and team planning.